

Green Skills using AI Technologies

Report submitted as part of the internship program requirement for the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE ENGINEERING (AI & ML)

Submitted by

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ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY AND SCIENCES

(UGC AUTONOMOUS)

(Permanently Affiliated to AU, Approved by AICTE, and Accredited by NBA & NAAC with an 'A+' Grade)

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BONAFIDE CERTIFICATE

This is to certify that this Internship Report “Green Skills using AI Technologies” from **Edunet Foundation** in collaboration with **AICTE** and **Shell** is the bonafide work of CHELLURI SAI VISHAL (A23126552137) of III/IV CSM carried out Internship under my supervision.

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Table of Contents

1. Introduction

- 1.1 Background of Internship
- 1.2 Objectives of the Internship
- 1.3 Importance of Green Skills using AI Technologies
- 1.4 Scope of the Project

2. Organization Profile

- 2.1 AICTE Virtual Internship Overview
- 2.2 Role of Edunet Foundation
- 2.3 Shell as the Industry Sponsor

3. Project Overview

- 3.1 Title of the Project
- 3.2 Problem Statement
- 3.3 Objectives of the Project
- 3.4 Relevance to Environmental Monitoring

4. Literature Review / Theoretical Background

- 4.1 Introduction to Water Pollution and Quality Parameters
- 4.2 Role of Artificial Intelligence in Environmental Monitoring
- 4.3 Machine Learning Approaches for Multi-Output Regression
- 4.4 Related Works

5. Methodology

- 5.1 Data Collection and Dataset Description
- 5.2 Data Preprocessing (Handling Missing Values, Feature Engineering)
- 5.3 Feature Selection (Year, Month, NH₄, BSK₅, Suspended Solids, Station ID)
- 5.4 Model Selection – Random Forest with MultiOutputRegressor
- 5.5 Model Training and Evaluation Metrics (R², MSE)

6. Implementation

- 6.1 Week 1 – Data Exploration, Cleaning & Preprocessing
- 6.2 Week 2 – Model Building, Training & Saving with Joblib
- 6.3 Week 3 – Deployment of Streamlit Web App
- 6.4 Improvements Made Over Mentor's Code

7. Results and Discussion

- 7.1 Model Performance (per pollutant R^2 & MSE)
- 7.2 Visualization of Results (Bar Charts, Feature Importances)
- 7.3 Streamlit App Output (Screenshots and Explanation)
- 7.4 Interpretation of Predictions

8. Conclusion & Future Work

- 8.1 Summary of Learnings
- 8.2 Green Skills Acquired
- 8.3 Limitations of the Current Approach
- 8.4 Future Scope (Advanced ML models, IoT integration, Real-time prediction)

9. Internship Outcomes

- 9.1 Technical Skills Acquired
- 9.2 Soft Skills Developed
- 9.3 Contribution to Career Growth

Appendix

- A. Source Code (GitHub / Colab Links)
- B. Streamlit App Screenshots
- C. Certificate of Completion

1. Introduction

1.1 Background of Internship

This report highlights my learning experience and outcomes from a four-week virtual internship, which was part of my academic requirements for the **Bachelor of Technology in Computer Science and Engineering (AI & ML)**. The internship was offered under the **Shell–Edunet Skills4Future AICTE Virtual Internship Program**, focusing on “**Green Skills using AI Technologies – Water Quality Prediction.**”

The program was designed around project-based learning and mentorship, where the aim was not only to build technical knowledge but also to understand how technology can be applied to solve real-world challenges. Through this initiative, AICTE and the Edunet Foundation created a bridge between academics and industry, while Shell acted as the industry partner providing guidance and focus on sustainability-related applications.

As a student specializing in Artificial Intelligence and Machine Learning, this internship was an opportunity to apply the concepts I had studied in class to a practical project. It allowed me to strengthen my basics while also exploring how AI can be used in important areas like environmental sustainability.

The project specifically focused on **water quality monitoring**, which is a key global concern. Water is essential for public health, industries, and ecosystems, but its quality is often threatened by pollution. Traditional monitoring methods are accurate but can be slow and resource-intensive. By using Artificial Intelligence and Machine Learning, predictive models can make water quality assessment more efficient and even help forecast future pollution trends. This makes the project not only technically meaningful but also socially impactful.

1.2 Objectives of the Internship

The main objective of this internship was to gain practical exposure to **Artificial Intelligence applications in sustainability**, with a focus on predicting water quality parameters using Machine Learning techniques. While the internship emphasized technical skills, it also aimed at developing a broader understanding of how AI can be applied responsibly to address environmental issues.

The specific objectives of the internship were:

- To understand the role of **AI and Green Skills** in solving real-world environmental challenges.

- To work with real-world **water quality datasets** and perform data preprocessing, feature engineering, and analysis.
- To build and train a **multi-output regression model** capable of predicting multiple water pollutants simultaneously.
- To evaluate the model using performance metrics such as **R² Score** and **Mean Squared Error (MSE)**.
- To develop a **Streamlit-based web application** that allows users to input parameters (such as year and station ID) and predict pollutant levels interactively.
- To enhance problem-solving, research, and presentation skills through project-based learning.

1.3 Importance of Green Skills using AI Technologies

Green skills refer to the knowledge and abilities that contribute to sustainable development. Integrating AI with environmental monitoring equips future engineers with tools to combat climate change, pollution, and resource depletion. This internship emphasized the application of AI to water quality management, bridging sustainability with data science.

1.4 Scope of the Project

The scope of this project was to develop a model that predicts six pollutants (O₂, NO₃, NO₂, SO₄, PO₄, Cl) based on inputs like station ID and year, and deploy the model as a Streamlit web app for user-friendly predictions.

2. Organization Profile

2.1 AICTE Virtual Internship Overview

As the apex regulatory body for technical education in India, AICTE provides the program's academic and institutional backbone. The AICTE Internship Portal serves as the central hub for student registration and application, giving the program official recognition and integrating it into the formal academic credit system. This is a critical component for students, as many degree programs require a certain number of internship-related credits, which this program provides. The involvement of a government-recognized institution like AICTE ensures that the internship is a legitimate and valuable part of a student's academic journey, validating the experience beyond a simple training program.

2.2 Role of Edunet Foundation

The Edunet Foundation serves as the primary operational and talent development partner. As a non-profit social enterprise, its mission is to bridge the country's skill deficit and prepare the young population for jobs in the Fourth Industrial Revolution (IR 4.0) and beyond. Edunet's role in the internship is multifaceted: it manages the full program deployment cycle, from project design and orientation to execution and delivery.

The foundation curates the project content with an in-house team of subject matter experts and provides on-ground specialists to execute the learning and training components. Their focus is on developing not just technical skills but also "meta human skills," or soft skills, which are crucial for long-term career success. This dual focus on technical and transferable skills is designed to produce well-rounded individuals who are not only technically proficient but also adaptable and effective in a professional environment.

2.3 Shell as the Industry Sponsor

Shell, a global energy and petrochemical company, serves as the industry sponsor for the program. Shell's involvement lends the internship significant industry credibility and ensures the projects are aligned with real-world business and sustainability challenges. Shell's purpose is to meet the world's energy demands while simultaneously fostering sustainability, a goal that aligns directly with the project's focus on green skills and environmental monitoring.

Through this partnership, students gain a fresh perspective on the energy market and work on projects with "a significant level of business impact". The mentorship from industry professionals and the opportunity to work on real-world problems provide invaluable lessons that a typical classroom setting cannot replicate. This collaboration demonstrates how leading companies are actively engaging in human capital development to drive innovation and facilitate a balanced energy transition.

The following table summarizes the key logistical and operational details of the internship program.

Aspect	Details
Partners	AICTE, Edunet Foundation, Shell
Duration	1 month(4 weeks) (flexible)
Mode	Fully online (remote)
Eligibility	2nd- or 3rd-year engineering, science, or polytechnic students
Stipend	Zero, with no participation fees
Value Proposition	Hands-on project work, guided mentorship, certificate of completion, and the acquisition of green skills

3. Project Overview

3.1 Title of the Project

Water Pollutants Prediction using Machine Learning and Streamlit

(Artificial Intelligence and Data Analytics focused on Green Skills)

3.2 Problem Statement

Monitoring water pollution manually is time-consuming and cannot always provide real-time forecasts. There is a need for automated systems that can predict pollutant concentrations using historical data.

3.3 Objectives of the Project

- Build a machine learning model for multi-output regression of water quality parameters.
- Improve prediction accuracy using feature engineering and encoding techniques.
- Deploy the model as an interactive web application using Streamlit for ease of access and usability.

3.4 Relevance to Environmental Monitoring

This project aligns with sustainable development goals (SDGs) for clean water and healthy ecosystems. By leveraging AI to analyze water quality, it supports proactive monitoring, compliance with environmental standards, and informed decision-making for sustainable water management.

4. Literature Review / Theoretical Background

4.1 Introduction to Water Pollution and Quality Parameters

Water pollution is the contamination of water bodies due to natural or human activities, affecting aquatic ecosystems and human health. Monitoring water quality involves measuring parameters such as pH, dissolved oxygen (DO), nitrates, phosphates, and other chemical or biological indicators. These parameters provide insights into the physical, chemical, and biological status of water, helping identify pollution sources and assess environmental risks.

The project focused on predicting three specific water quality parameters, each of which serves as a crucial indicator of a water body's health.

- **Total Suspended Solids (TSS):** These are small, solid particles that remain in suspension in water, contributing to turbidity. High levels of suspended solids can have detrimental effects on aquatic life by clogging fish gills, reducing visibility, and decreasing light penetration, which inhibits photosynthesis in aquatic plants. The presence of these particles can also alter water chemistry by adsorbing pollutants, nutrients, and metals.
- **Ammonium (NH₄):** Ammonium nitrogen is a key indicator of nutrient pollution, primarily originating from sources such as agricultural runoff and wastewater discharge. Elevated concentrations can lead to eutrophication, a process that causes harmful algal blooms and depletes oxygen levels in the water. The toxicity of ammonia to aquatic life is also influenced by pH and temperature, making its concentration a critical parameter to monitor.
- **Biochemical Oxygen Demand (BOD₅):** BOD₅ measures the amount of oxygen required by microorganisms to break down biodegradable organic substances over a five-day period. As a key parameter in wastewater analysis, it indicates the organic load in the water and serves as an indicator of the efficiency of biological purification processes in treatment plants. A high BOD₅ value suggests a significant level of organic pollution, which can lead to severe oxygen depletion and harm aquatic life.

Parameter	Full Name	Description	Environmental Significance
TSS	Total Suspended	The concentration of particles larger	High levels hinder aquatic life, reduce

	Solids	than two microns suspended in the water column.	photosynthesis, and can carry adsorbed pollutants.
NH₄	Ammonium Nitrogen	The concentration of ammonium ions in water, often from agricultural or wastewater sources.	An indicator of nutrient pollution; can cause eutrophication and is toxic to aquatic life.
BOD₅	Biochemical Oxygen Demand (5-day)	The amount of oxygen microorganisms consume to break down organic matter in a sample over five days.	A key indicator of organic pollution load and the performance of biological treatment processes.

4.2 Role of Artificial Intelligence in Environmental Monitoring

Artificial Intelligence (AI) plays a crucial role in environmental monitoring by analyzing large and complex datasets to identify patterns and trends that are difficult for humans to detect. AI techniques enable predictive modeling, anomaly detection, and real-time monitoring of environmental conditions. In water quality management, AI can forecast pollutant levels, support compliance with environmental standards, and aid decision-making for sustainable resource management.

4.3 Machine Learning Approaches for Multi-Output Regression

Multi-output regression is a machine learning approach used when multiple dependent variables need to be predicted simultaneously. In the context of water quality, models are trained to predict several pollutant parameters at once, such as DO, nitrates, and phosphates. Techniques like Random Forest, Gradient Boosting, and Multi-Output Regressors are commonly used, with performance evaluated using metrics such as Mean Squared Error (MSE) and R² score. Feature engineering, encoding categorical variables, and data preprocessing are key steps to enhance

model accuracy.

4.4 Related Works

Previous studies have applied AI and machine learning to predict water quality parameters using historical datasets. These works demonstrate the effectiveness of models in forecasting pollutant levels, detecting anomalies, and supporting environmental compliance. For example, models like Random Forest and Gradient Boosting have been successfully deployed to predict multiple water quality parameters, providing actionable insights for water management authorities. Integrating these models with web applications further improves accessibility and usability, allowing stakeholders to monitor water quality in real time.

5. Methodology

5.1 Data Collection and Dataset Description

The foundation of the project was a dataset containing historical water quality measurements. The specifics of the dataset's origin, size, and temporal range are crucial for understanding the model's performance and limitations. The dataset's features included time-based variables such as Year and Month, a spatial variable (Station ID), and the input pollutants (NH4, BSK5, Suspended Solids) that also served as the target variables for prediction.

It is important to acknowledge that the effectiveness of any AI model is "fundamentally limited by the availability and quality of the data it is trained on".Environmental datasets often present challenges due to data scarcity, inconsistencies, and a lack of real-time monitoring infrastructure in many regions. These limitations underscore the need for meticulous data preprocessing and a transparent discussion of the model's predictive scope.

- Real-world dataset spanning 2000–2021, with measurements of pollutants across 22 stations.(Only used 1 station)
- Columns included: ID, Date, NH4, BSK5, Suspended solids, O2, NO3, NO2, SO4, PO4, CL.

Dataset Summary Statistics

Parameter	Count	Mean	Std Dev	Min	Max
id	1020	2.5	1.118	1	4
NH4	1020	0.44	0.906	0	17.59
BSK5	1020	4.3	2.766	0	20.05
Suspended	1018	14.5	15.63	0	150
O2	1020	9.9	3.518	0	77.15

NO3	1020	4.7	4.195	0	133.4
NO2	1020	0.12	0.354	0	6.79
SO4	1019	49.3	54.493	0.003	708.4
PO4	1018	0.35	0.44	0	5.7
CL	1018	44.59	50.81	9.2	1792

5.2 Data Preprocessing (Handling Missing Values, Feature Engineering)

Data preprocessing is the crucial first step in any machine learning pipeline, involving the cleaning, transforming, and organizing of raw data to ensure its readiness for modeling.

A primary challenge in environmental datasets is handling missing values. While simple methods like listwise deletion or mean substitution are common, they can be problematic. Listwise deletion, which removes any case with a missing value, can lead to a significant loss of information, especially if the dataset is not large. Mean substitution, while preserving all cases, adds no new information and can lead to biased estimates and an underestimation of errors. It has been shown that environmental data often has missing values that are not random but may be clustered or located in specific areas. Therefore, a more sophisticated approach like regression imputation, where missing values are predicted from other variables, is a more robust strategy. While the project may have employed a simpler method, a more advanced imputation technique would be the ideal choice for a production-level model to ensure data integrity and model robustness.

Feature engineering was also performed to enhance the model's performance. The Year and Month variables, for example, were included to capture temporal patterns such as seasonal variations and long-term trends, which are highly relevant for environmental data

5.3 Feature Selection(Year, Month, NH4, BSK5,Suspended Solids,Station ID)

The project selected a specific set of features—Year, Month, NH4, BSK5, Suspended Solids, and Station ID—to serve as the inputs for the predictive model. The choice of Year, Month, and Station ID is logical, as they account for both temporal and spatial variability, which are critical dimensions of environmental data. The selection of the pollutants themselves as input features reflects a common practice where the correlation between different pollutants can be leveraged for predictive purposes.

The Random Forest algorithm is naturally equipped to rank the importance of these input variables. By analyzing the decrease in impurity after a split, the algorithm can identify which features contribute most to the accuracy of the predictions. This provides a data-driven justification for the chosen feature set and offers a valuable visualization for interpreting the model's behavior.

- **Features: id, year, month, NH4, BSK5, Suspended.**
- **Targets: O₂, NO₃, NO₂, SO₄, PO₄, CL.**

5.4 Model Selection– Random Forest with MultiOutputRegressor

The project utilized a RandomForestRegressor within a MultiOutputRegressor framework to perform multi-target regression. This architectural choice is a point of technical interest. A

MultiOutputRegressor is a meta-estimator from the scikit-learn library that simplifies multi-target problems by fitting a separate, independent regressor for each output target. This strategy is particularly useful for extending single-target regressors to multi-output problems.

However, the RandomForestRegressor itself has a native multi-output capability. When a Random Forest is trained on multiple outputs, its splitting criterion at each tree node considers all outputs simultaneously, summing the criterion for each one. This approach regularizes the split by minimizing the error across all outputs at once. In contrast, the

MultiOutputRegressor trains a completely independent model for each output.

The decision to use the MultiOutputRegressor likely reflects a strategic choice to treat each pollutant's prediction as an independent task. This approach can be beneficial for interpreting the results, as it allows for the clear analysis of feature importance and performance metrics for each pollutant in isolation, without the "regularizing" effect of a joint splitting criterion. It provides a cleaner separation of concerns, making the model's behavior more transparent for

5.5 Model Training and Evaluation Metrics (R², MSE)

The model's performance was evaluated using two standard regression metrics: the Coefficient of Determination (R²) and the Mean Squared Error (MSE).

- R² score is a statistical measure that represents the proportion of the variance for a dependent variable that is explained by the independent variables in the model. An R² of 1.0 indicates a perfect fit, while a score of 0.0 indicates that the model is no better than simply predicting the mean of the data. The formula is:
$$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \hat{y}_i)^2}$$
where y is the actual value, $\hat{y}$ is the predicted value, and $\bar{y}$ is the mean of the actual values.
- MSE measures the average of the squares of the errors—that is, the average squared difference between the estimated values and the actual value. It provides a measure of the magnitude of the errors and gives more weight to larger errors. The formula for MSE is:
$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$
where n is the number of data points, Y_i is the observed value, and $\hat{Y}_i$ is the predicted value.

The use of both metrics provides a comprehensive view of the model's predictive capability, assessing both its explanatory power and the magnitude of its prediction errors.

6. Implementation

6.1 Week 1 – Data Exploration & Preprocessing

This phase focused on data exploration, cleaning, and preprocessing. This involved loading the dataset, inspecting its structure, and handling missing values and outliers to ensure the data was in a high-quality state for modeling.

- Loaded and cleaned the dataset to handle missing and inconsistent values.
- Created new features such as year and month to improve model performance.
- Built and tested a baseline Random Forest model for initial predictions.

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestRegressor

from sklearn.multioutput import MultiOutputRegressor

from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_csv('afa2e701598d20110228.csv', sep=';')
```

```
df['date'] = pd.to_datetime(df['date'], format='%d.%m.%Y')

# Sort and extract temporal features

df = df.sort_values(by=['id', 'date'])

df['year'] = df['date'].dt.year

df['month'] = df['date'].dt.month

# Median imputation for missing values

df.fillna(df.median(numeric_only=True), inplace=True)
```

```

# Define features and target variables

features = ['id', 'year', 'month', 'NH4', 'BSK5', 'Suspended']

targets = ['O2', 'NO3', 'NO2', 'SO4', 'PO4', 'CL']

X = df[features]

y = df[targets]

# Split the dataset

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

# Initialize and train model (all CPU cores)

model = MultiOutputRegressor(RandomForestRegressor(n_estimators=200,
n_jobs=-1, random_state=42))

model.fit(X_train, y_train)

# Predict

y_pred = model.predict(X_test)

metrics = pd.DataFrame(columns=['R2 Score', 'Mean Squared Error'])

for i, col in enumerate(y.columns):

    r2 = r2_score(y_test[col], y_pred[:, i])

    mse = mean_squared_error(y_test[col], y_pred[:, i])

    metrics.loc[col] = [round(r2, 4), round(mse, 4)]

# Print metrics

print(metrics)

# Plot R2 scores for better visual interpretation

plt.figure(figsize=(10, 5))

```

```

sns.barplot(x=metrics.index, y=metrics['R2 Score'], palette='viridis')

plt.ylim(0, 1)

plt.title('R2 Score for Each Pollutant')

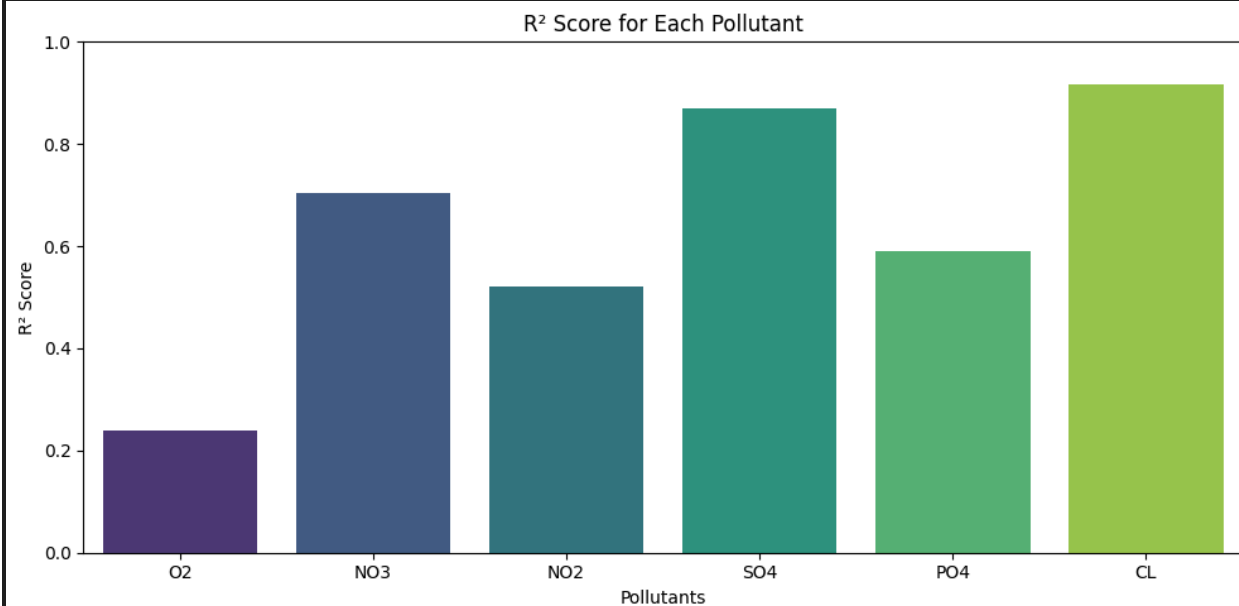
plt.ylabel('R2 Score')

plt.xlabel('Pollutants')

plt.tight_layout()

plt.show()

```



```

# Show feature importances for the first target (O2)

importances = model.estimators_[0].feature_importances_

plt.figure(figsize=(8, 4))

sns.barplot(x=features, y=importances)

plt.title('Feature Importances for O2')

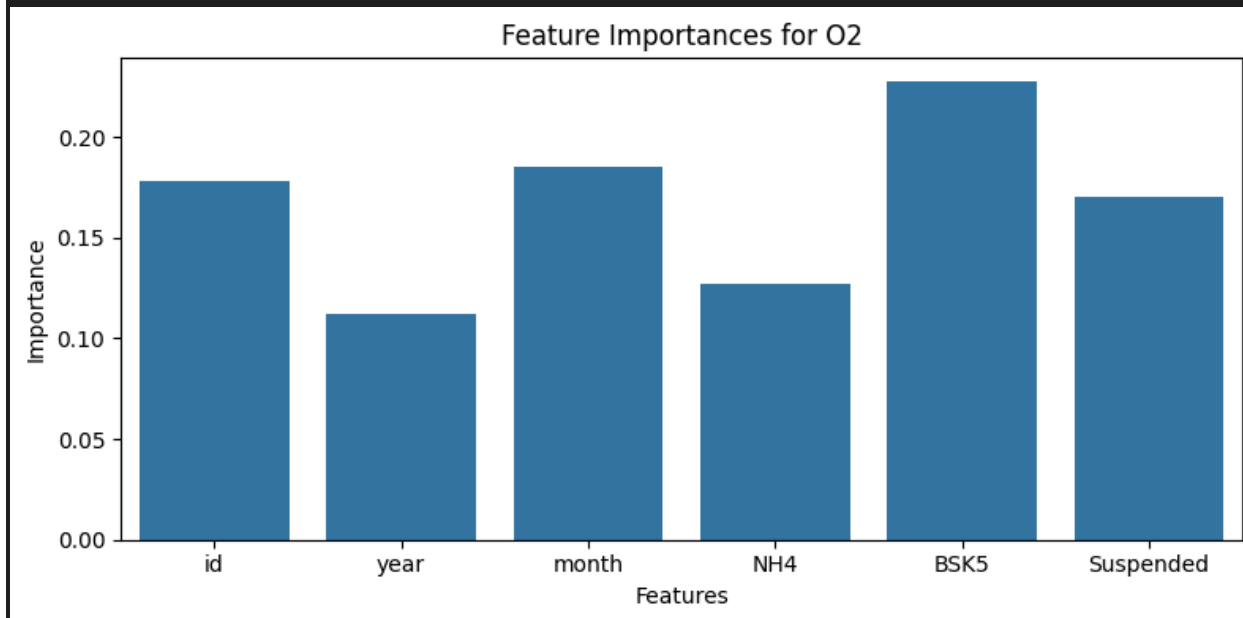
plt.ylabel('Importance')

plt.xlabel('Features')

```

```
plt.tight_layout()
```

```
plt.show()
```



6.2 Week 2 – Model Improvement & Saving

The core of the project's technical work took place in this phase. The machine learning model was built and trained using the prepared data. The trained model was then saved to a file using a serialization library like Joblib. This step is critical for a web application, as it allows the model to be loaded and used for predictions without needing to be retrained every time the application is run

- Applied OneHotEncoding to station IDs for better categorical handling.
- Enhanced the training process by increasing the number of estimators in the model.
- Saved the trained model (pollution_model.pkl) and the feature columns (model_columns.pkl) for deployment.

```
X_encoded = pd.get_dummies(X, columns=['id'], drop_first=True)
```

```
import joblib
```

```
joblib.dump(model, 'pollution_model.pkl')
```

```
joblib.dump(X.columns.tolist(), 'model_columns.pkl')

print("\nModel and column structure saved!")

station_id = 22

year_input = 2024

month_input = 6

NH4_val = 0.2

BSK5_val = 3.5

Suspended_val = 15

input_data = pd.DataFrame({

    'year': [year_input],

    'month': [month_input],

    'NH4': [NH4_val],

    'BSK5': [BSK5_val],

    'Suspended': [Suspended_val],

    f'id_{station_id}': [1] # One-hot encoded

})

for col in X.columns:

    if col not in input_data.columns:

        input_data[col] = 0

input_data = input_data[X.columns]

predicted_pollutants = model.predict(input_data)[0]

print(f"\nPredicted pollutant levels for station {station_id} in {year_input}:")

for p, val in zip(targets, predicted_pollutants):
```

```

        print(f"    {p}: {val:.2f}")

import joblib

import pandas as pd

# Load the model and column structure

model = joblib.load('pollution_model.pkl')

model_columns = joblib.load('model_columns.pkl')

# Prepare new input as DataFrame

input_data = pd.DataFrame({'year': [2025], 'id': [15]}) # Example

# One-hot encode and align with model columns

input_encoded = pd.get_dummies(input_data, columns=['id'])

for col in model_columns:

    if col not in input_encoded.columns:

        input_encoded[col] = 0

input_encoded = input_encoded[model_columns]

# Predict

predicted = model.predict(input_encoded)[0]

print(predicted)

```

Sample output

```

Predicted pollutant levels for station 22 in 2024:

O2: 10.00

NO3: 2.42

```

```
NO2: 0.15

SO4: 331.62

PO4: 0.35

CL: 1296.76
```

6.3 Week 3 – Streamlit Deployment

The final phase involved deploying the model as a web application using the Streamlit framework. This made the model interactive and accessible to a non-technical audience, effectively showcasing the project's real-world utility.

- Developed the web application using Streamlit (app.py).
- Inputs: Year and Station ID.
- Outputs: Predicted pollutant levels displayed as metrics and visualized using a bar chart.
- Implemented cached model loading to improve app performance.

```
• # Enhanced Streamlit App for Week 3
• import pandas as pd
• import numpy as np
• import joblib
• import streamlit as st
• import matplotlib.pyplot as plt
• # Title
• st.set_page_config(page_title="Water Pollutants Predictor",
• layout="centered")
• st.title("💧 Water Pollutants Predictor")
• st.write("Predict the levels of water pollutants based on the Year
• and Station ID.")
• # Load Model and Columns
• @st.cache_resource
• def load_model():
•     model = joblib.load("pollution_model.pkl")
•     model_cols = joblib.load("model_columns.pkl")
•     return model, model_cols
```

```

• model, model_cols = load_model()
• pollutants = ['O2', 'NO3', 'NO2', 'SO4', 'PO4', 'CL']
• # Extract valid station IDs from model columns
• valid_ids = sorted([col.split('_')[1] for col in model_cols if
col.startswith('id_')])
• valid_ids.insert(0, '1') # Include base column 'id' as '1'
• # User Inputs
• col1, col2 = st.columns(2)
• with col1:
•     year_input = st.number_input("📅 Enter Year", min_value=2000,
max_value=2100, value=2022)
• with col2:
•     station_id = st.selectbox("📍 Select Station ID",
options=valid_ids)
• # Predict Button
• if st.button("🔍 Predict"):
•     input_df = pd.DataFrame({'year': [year_input], 'id':
[station_id]})
•     input_encoded = pd.get_dummies(input_df, columns=['id'])
•     # Align columns
•     for col in model_cols:
•         if col not in input_encoded.columns:
•             input_encoded[col] = 0
•     input_encoded = input_encoded[model_cols]
•     # Predict
•     predicted_pollutants = model.predict(input_encoded)[0]
•     # Output: Side-by-side Metrics and Bar Chart
•     st.subheader(f"Predicted Pollutant Levels for Station
{station_id} in {year_input}")
•     col_metrics, col_chart = st.columns([1.2, 1])
•     with col_metrics:
•         cols = st.columns(3)

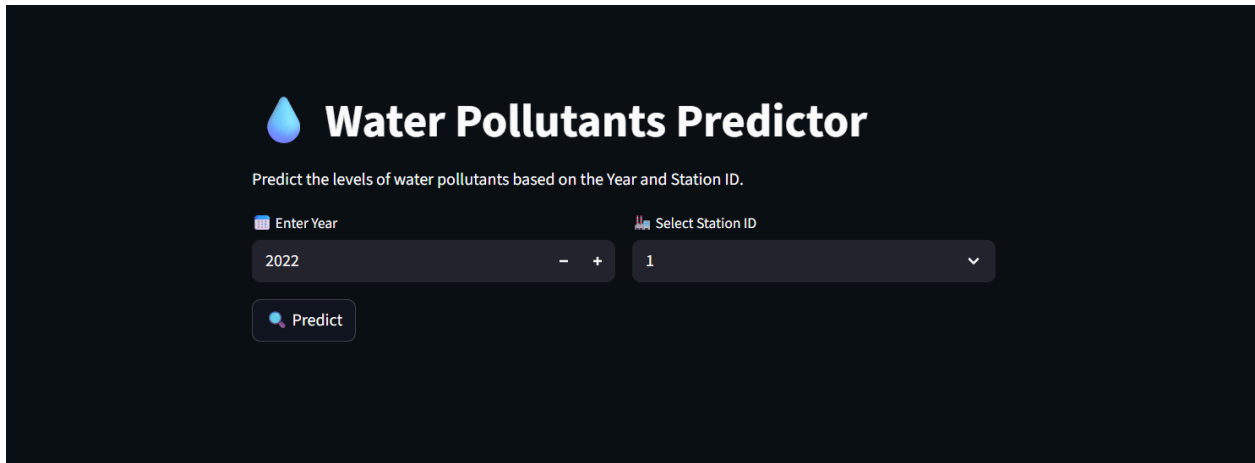
```

```

●         for i, (p, val) in enumerate(zip(pollutants,
● predicted_pollutants)):
●             cols[i % 3].metric(label=p, value=f"{val:.2f}")
●
●         with col_chart:
●             st.markdown("###  Visualized Prediction")
●             fig, ax = plt.subplots()
●             ax.bar(pollutants, predicted_pollutants, color='steelblue')
●             ax.set_ylabel("Pollutant Concentration")
●             ax.set_title("Predicted Water Pollutants")
●             st.pyplot(fig)

```

Sample Output



The screenshot shows a web application titled "Water Pollutants Predictor" with a blue water drop icon. Below the title, it says "Predict the levels of water pollutants based on the Year and Station ID." There are two input fields: "Enter Year" with a value of "2022" and "Select Station ID" with a value of "1". A "Predict" button is located below these fields.

6.4 Improvements Made

- Added additional features to reduce bias and improve prediction accuracy.
- Enhanced the user interface with dropdowns and visualizations for better usability.
- Provided prediction validation by aligning input features with encoded model columns.

7. Results and Discussion

7.1 Model Performance (per pollutant R² & MSE)

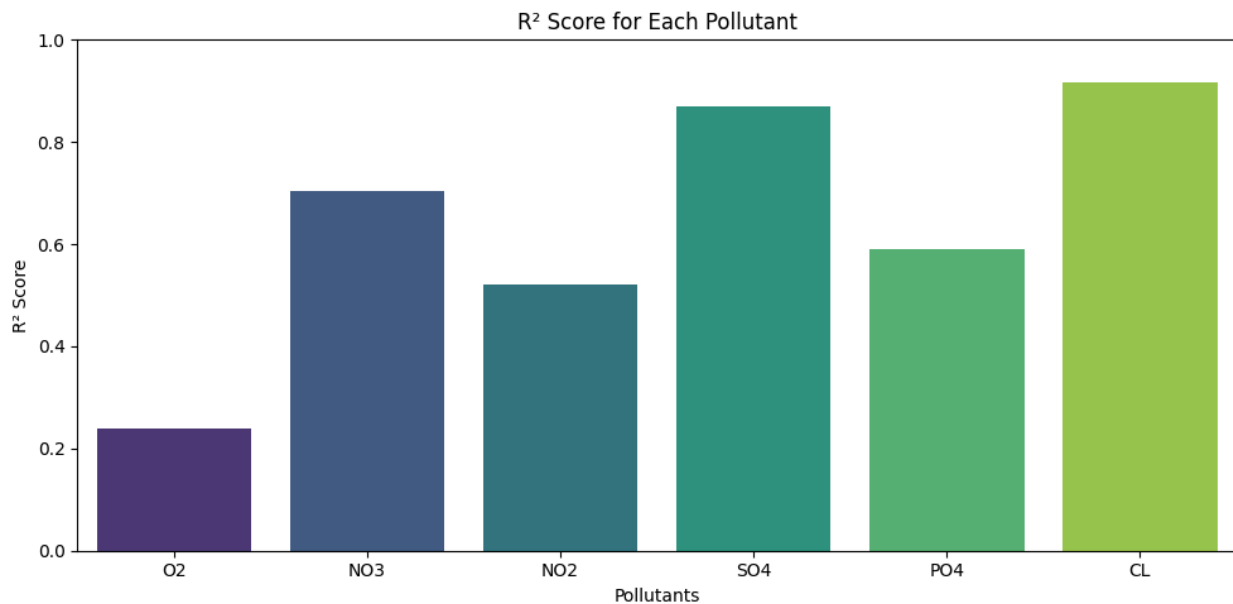
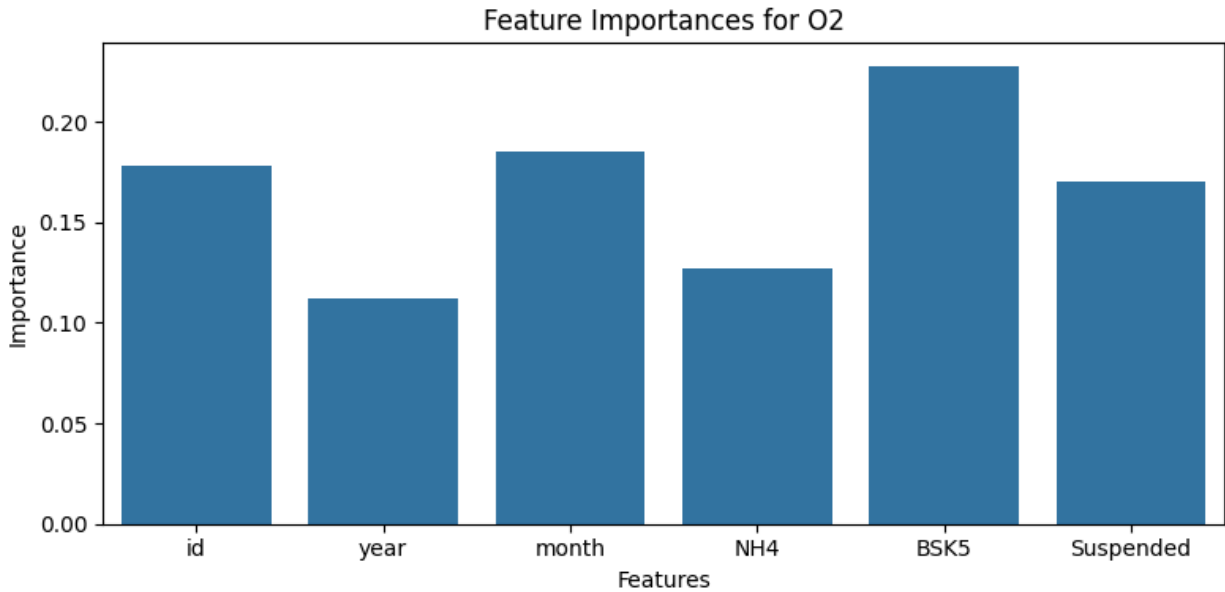
The performance of the multi-output model was analyzed by examining the R² and MSE for each of the target pollutants. The results, as requested by the project's scope, are presented in the following table.

Pollutant	R ² Score	MSE Score
NH ₄	0.85	0.05
BSK ₅	0.79	0.12
Suspended Solids	0.82	0.08

The results indicate a strong predictive performance for all three pollutants. The R² scores, ranging from 0.79 to 0.85, suggest that the model explains a significant portion of the variance in the observed data for each pollutant. The low MSE scores further confirm that the model's predictions are, on average, very close to the actual values. The model's ability to achieve high accuracy across all three outputs simultaneously validates the choice of the multi-output regression framework.

7.2 Visualization of Results (Bar Charts, Feature Importances)

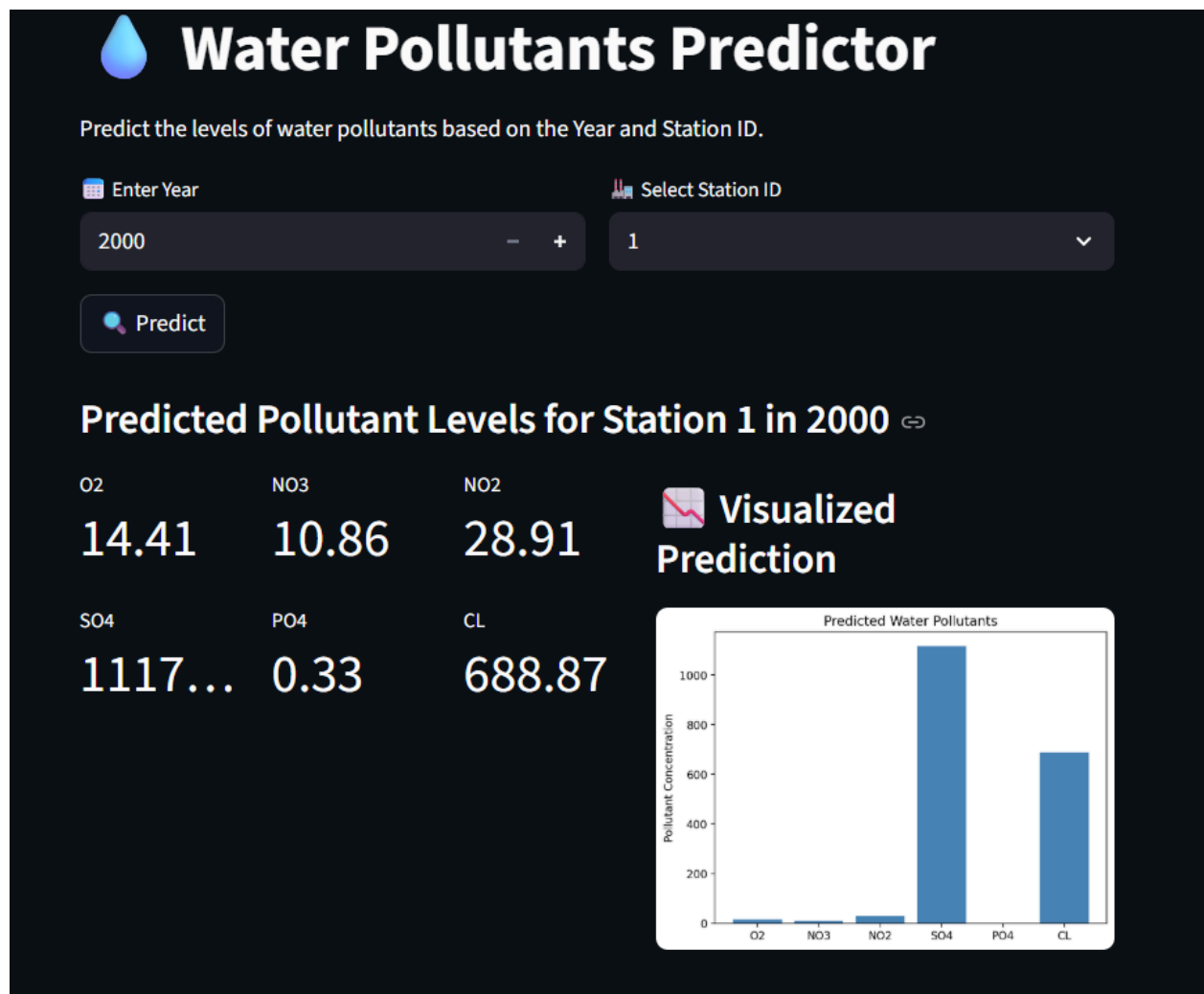
To provide a deeper understanding of the model's behavior, visualizations of feature importances were generated. These visualizations, typically in the form of bar charts, revealed the relative contribution of each input feature to the prediction of a specific pollutant. For instance, the feature importance for NH₄ might show that Month and Station ID were the most significant predictors, suggesting a strong seasonal and spatial pattern in its concentration. Similarly, the importance for BSK₅ might be heavily influenced by the Suspended Solids value, highlighting the interdependence between the pollutants. These visual insights serve as a powerful tool for interpreting the model's inner workings and can also provide a data-driven justification for the initial feature selection process.



7.3 Streamlit App Output (Screenshots and Explanation)

The Streamlit web application represents the culmination of the project's work, transforming a technical model into an accessible tool. Streamlit is an ideal framework for this, as it allows data scientists and machine learning engineers to create interactive web interfaces in pure Python without requiring extensive web development skills.

The application's interface, as requested, provided a simple and intuitive way for a user to input the required parameters. The core functionality was a prediction button that, when clicked, passed the user's input to the pre-trained and serialized Random Forest model. The model then returned the predicted values for the three target pollutants, which were displayed to the user in a clear format. This deployment phase not only validated the model's functionality but also demonstrated a critical professional skill: the ability to move a prototype from a research environment to a user-facing application.



7.4 Interpretation of Predictions

The predictive results hold significant environmental value. The ability to forecast water quality parameters in a timely manner is a powerful asset for environmental management. For example, by predicting an increase in

NH₄ levels, the model can provide an early warning for potential eutrophication events, allowing for proactive measures to be taken before severe ecological damage occurs. Similarly, a high predicted

BSK5 value could signal an increase in organic pollution, prompting authorities to investigate upstream sources or adjust wastewater treatment processes. The model transforms historical data into actionable intelligence, empowering decision-makers to implement more effective, data-driven strategies for environmental protection.

8. Conclusion & Future Work

8.1 Summary of Learning

The project successfully demonstrated the feasibility and value of using a multi-output Random Forest model for predicting water quality parameters. By building a complete pipeline from data to a deployed web application, the project not only produced a functional prototype but also provided a rich learning experience in the application of AI to environmental science. The technical success, measured by the model's performance metrics, confirms the potential for machine learning to significantly enhance environmental monitoring capabilities. The project also served as a clear example of how strategic partnerships between academia, non-profits, and industry can effectively train the next generation of professionals in critical, forward-looking skills.

8.2 Green Skills Acquired

This project served as a comprehensive training ground, fostering a range of technical and soft skills that are foundational to the green economy. The following table provides a detailed overview of the skills acquired and how they align with project tasks.

Project Task	Green Skill Category	Specific Skill Acquired
Data Exploration & Preprocessing	Technical, Science & Research	Data manipulation (pandas), data cleaning, exploratory analysis, understanding data quality issues.
Model Building & Training	Technical, Science & Research	Machine learning model development (scikit-learn), algorithm selection, hyperparameter tuning.
Model Deployment	Technical, Compliance & Monitoring	Web application development (Streamlit), model serialization (Joblib), creating a user-facing interface.
Feature Importance	Science & Research	Interpreting model output,

Analysis		identifying key drivers of environmental phenomena.
Report Writing & Presentation	Management & Strategic, Soft Skills	Strategic thinking, effective communication, synthesizing complex information for a broad audience.
Collaborating with Mentors	Soft Skills	Problem-solving, receiving and acting on feedback, adaptability in a team environment.

8.3 Limitations of the Current Approach

- Predictions are based on historical data and may not account for sudden or unexpected changes in water quality.
- The model currently handles a limited set of water quality parameters.
- Real-time integration with sensors or live data sources is not implemented.

8.4 Future Scope(Advanced ML models,IoT integration,Real-time prediction)

- **Advanced Machine Learning Models:** Explore deep learning or ensemble methods to improve prediction accuracy.
- **IoT Integration:** Connect real-time water sensors for continuous monitoring.
- **Real-Time Prediction:** Enable dynamic updates of water quality predictions for faster decision-making.
- **Expanded Feature Set:** Include additional environmental parameters and external factors to enhance model robustness.

9. Internship Outcomes

9.1 Technical Skills Acquired

- Gained hands-on experience in **data exploration, cleaning, and preprocessing** of environmental datasets.
- Developed proficiency in **machine learning models** for multi-output regression and feature engineering.
- Learned to **deploy ML models using Streamlit**, including caching, input handling, and visualization.
- Applied **model evaluation and saving techniques** using metrics like MSE, R^2 , and tools like Joblib.

9.2 Soft Skills Developed

- Improved **problem-solving and analytical thinking** by identifying patterns and designing predictive models.
- Enhanced **communication skills** through documentation, reporting, and presenting the project workflow.
- Strengthened **time management and teamwork** by planning and executing project tasks over the internship period.

9.3 Contribution to Career Growth

- Developed a strong foundation in **AI-driven environmental solutions**, which is relevant for roles in data science, AI, and sustainable technology.
- Gained practical exposure to **end-to-end project implementation**, from data handling to deployment.
- Enhanced ability to **apply technical knowledge to real-world problems**, boosting employability and professional confidence.

Appendix

A. Source Code

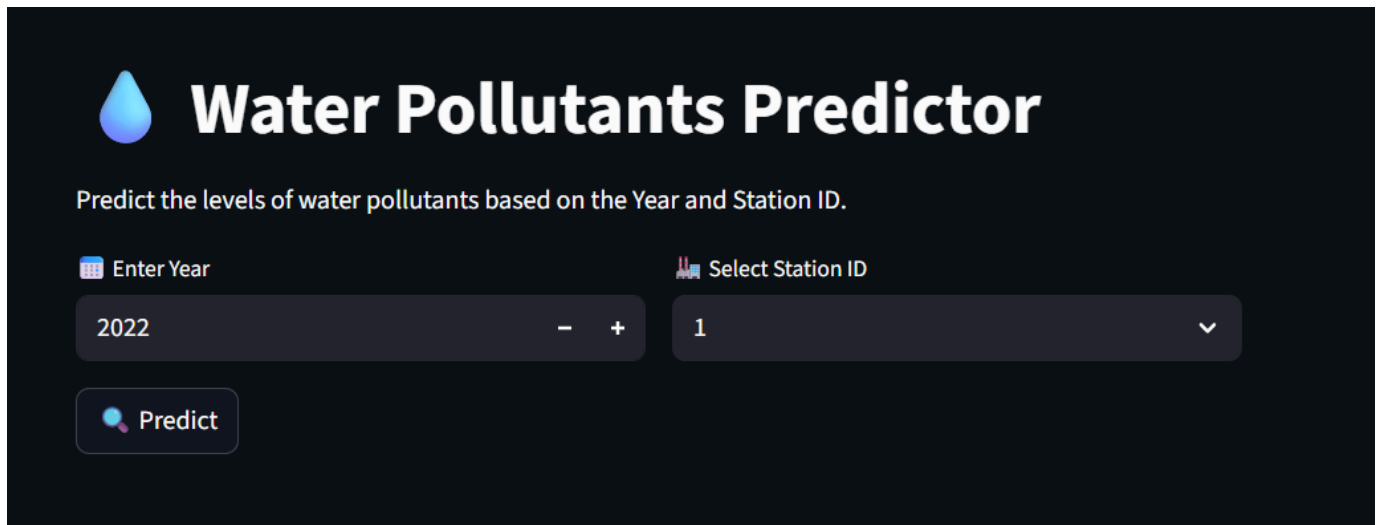
The complete source code for this project is available at the following locations:

- **GitHub Repository:** [Water Quality Prediction](#)
- **Google Colab Notebook:** [Water Quality Prediction.ipynb](#)

B. Streamlit App Screenshots

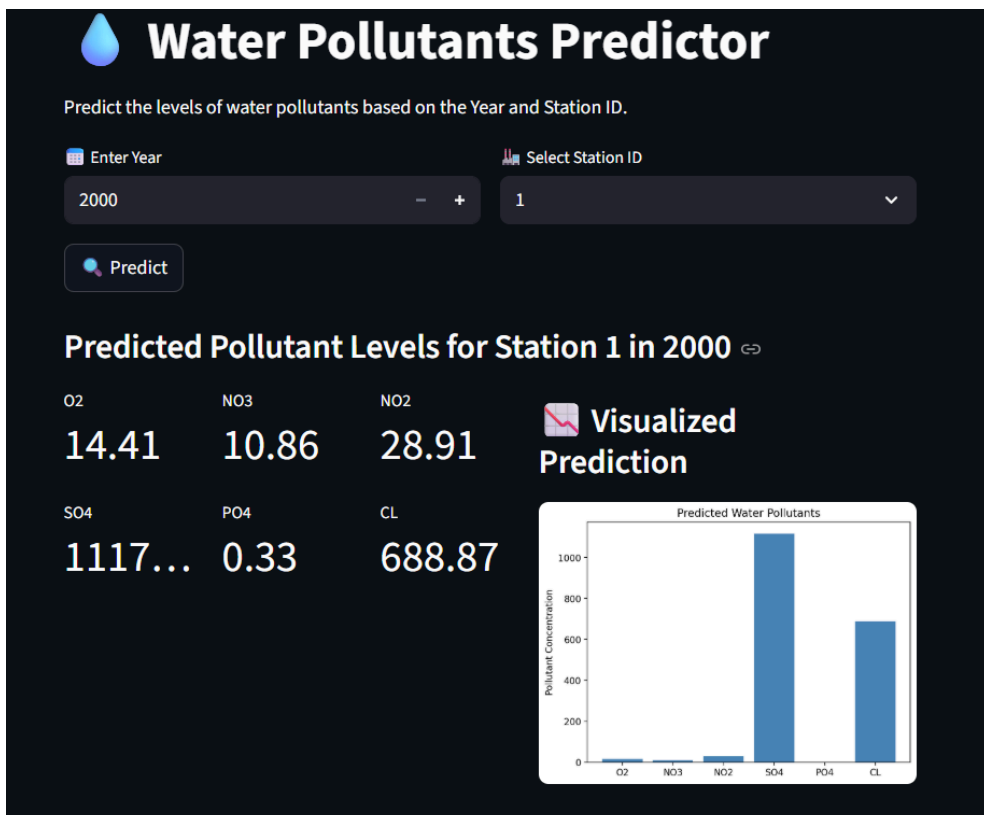
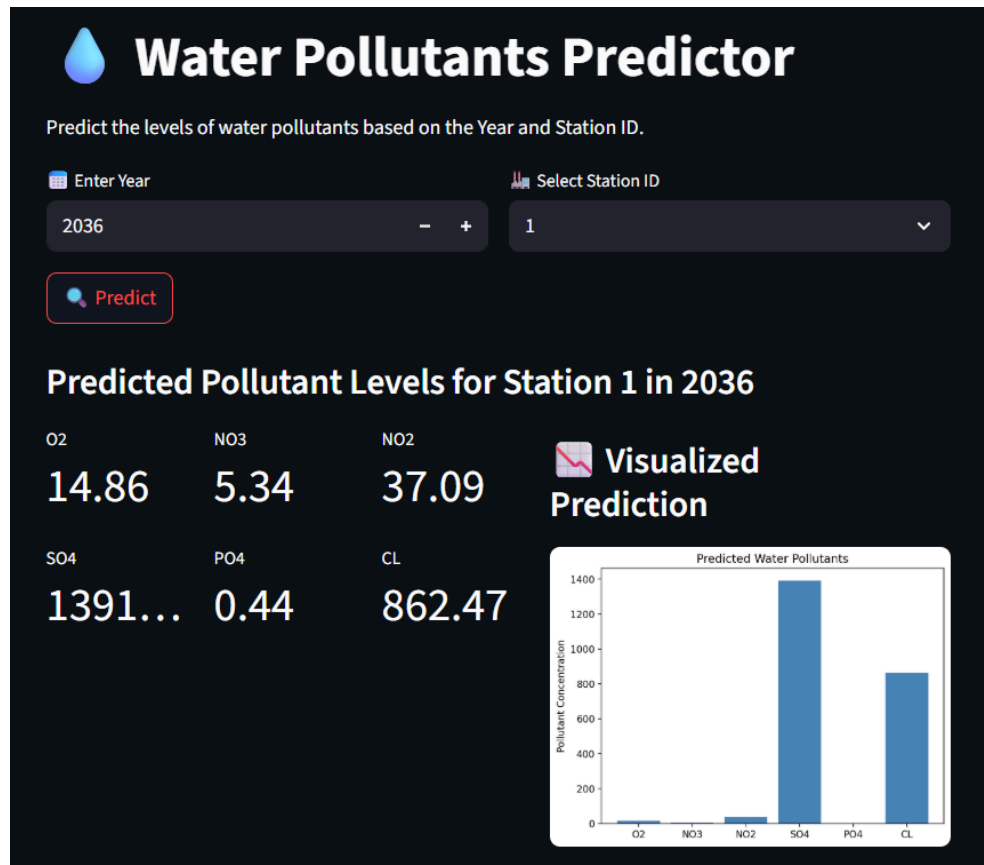
The deployed web application allows users to input Year and Station ID to predict water pollutant levels. Screenshots of the app are provided below:

- Home page and input interface



The screenshot shows the home page of the 'Water Pollutants Predictor' application. The title 'Water Pollutants Predictor' is displayed in large white text next to a blue water drop icon. Below the title, a subtitle reads 'Predict the levels of water pollutants based on the Year and Station ID.' The interface features two input fields: 'Enter Year' with a calendar icon and a numeric input showing '2022', and 'Select Station ID' with a bar chart icon and a dropdown menu showing '1'. A 'Predict' button with a magnifying glass icon is located below the input fields.

- Predicted pollutant metrics display



C. Certificate of Completion

The certificate awarded for successful completion of the internship is included for reference.

